

DIGITAL ANCESTRY: A SEMANTIC-AWARE RECOVERY FRAMEWORK FOR AI-EDITED IMAGES USING DEEP LEARNING DECODERS

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A Research Project

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CHAPTER 1

INTRODUCTION

1.1 Chapter Overview

This chapter lays the groundwork for the rest of the thesis. It begins with what has actually happened in the AI generated text space over the last three years, then narrows that situation into a formal problem statement split into a general dimension and a discipline specific dimension. From there, the chapter states the single research question this project is built to answer, together with four supporting sub questions, and explains the motivation that led the researcher toward this particular gap rather than any of the other directions available within AI-text forensics. The research aim and four research objectives follow, written in the identify analyse design evaluate format expected at undergraduate level. A rich picture then sets out how the proposed solution actually moves data through collection, rewriting, feature extraction, classification, and feature-survival analysis, before the chapter closes with the hardware and software resources committed to the project, the boundaries placed on its scope, and a short summary that carries the reader into Chapter 2.

1.2 Problem Background

By the start of 2026, four large language models had become the default writing assistants for university students almost everywhere: ChatGPT from OpenAI, Claude from Anthropic, Gemini from Google, and Grok from xAI. Each one is reachable through a free or low cost interface, and each is capable of producing a polished, submission ready paragraph in under ten seconds. Universities responded the way most institutions respond to a new risk they bought or built detection software. NSBM, like most universities, now routes submitted coursework through an AI detection layer before a human marker opens the file.

Almost every detector built so far answers one question: is this text AI generated, or is it not? That binary framing made sense in 2023, when ChatGPT was effectively the only model anyone needed to worry about. It makes far less sense now. A plagiarism office that flags a submission simply as “AI generated,” with nothing further attached, cannot tell whether a student used Grok once for a single paragraph or relied on ChatGPT across six consecutive assignments. Repeat-offence cases, appeals, and

formal misconduct hearings all depend on more than a flag. They depend on evidence that traces back to a specific tool, ideally across multiple submissions from the same student over time.

Two further developments make the binary approach even harder to defend. The first is that Grok, despite sitting alongside the other three as one of the most used commercial models by late 2025, does not appear in a single major detection or attribution benchmark not RAID (Dugan et al., 2024), not M4 (Wang et al., 2024), not DAIGT. The second is that students rarely submit raw model output anymore. A common workflow by late 2025 involves generating a first draft with one model, then feeding that draft into a second prompt something like “rewrite this so it sounds like me” before submission. That second step, LLM based rewriting, restructures sentences, swaps vocabulary, and reorganises paragraphs in ways a simple synonym substitution tool never could. Whether the forensic fingerprint left by the first model survives that second step has never actually been measured. That is the situation this study walks into.

1.3 Problem Statement

The shortfall described above is not a vague sense that detection “could be better.” It is measurable, and the numbers are not encouraging. Krishna et al. (2023) showed that DetectGPT’s accuracy at a fixed 1% false positive rate fell from 70.3% to 4.6% after a single round of DIPPER paraphrasing a drop of more than sixty-five percentage points from one editing step. Dugan et al. (2024) tested eleven detectors across eleven adversarial conditions in the RAID benchmark and found that five detectors lost an average of 40.6% accuracy under homoglyph substitution alone, with paraphrasing attacks ranking among the most damaging conditions tested across the board. Zhang et al. (2025) reported that targeted adversarial paraphrasing strips 30 to 50 percentage points from detection accuracy across several classifier architectures. None of these figures even describe attribution they describe whether a system can tell AI written text from human written text at all, and that narrower task already collapses under light editing pressure.

The Grok problem compounds the picture rather than sitting separately from it. Ye et al. (2025) tested classifiers trained on ChatGPT, Claude, and Gemini against Grok (AI) generated text and found zero agreement a model trained without Grok in its distribution could not classify Grok output any better than chance. Cao et al. (2025) documented a related homogenising effect across commercial LLMs, in which overlapping training corpora and similar RLHF tuning pipelines are pushing model outputs toward a shared stylistic centre, making attribution harder with every new model generation rather than easier. Put together: detection is fragile, attribution is a harder problem than detection to begin with, Grok is

invisible to the field, and nobody has checked what happens to attribution specifically once a second model rewrites the first model's output.

1.3.1 General Problem

At a general level, this leaves every institution relying on AI detection software holding a tool that gives a partial answer to the wrong question. Knowing that a submission is “AI generated,” with no further detail, offers little support to an academic integrity panel trying to establish a pattern of behaviour, distinguish a single lapse from repeated misuse, or defend a decision under appeal. The arms race between detection and evasion keeps pushing universities toward more expensive tooling while delivering systems that lose most of their power the moment a student adds one extra step a rewrite to their workflow. Institutional confidence in AI-detection infrastructure is currently being built on evidence that does not reflect how students actually use these tools in 2025 and 2026.

1.3.2 Specific Problem

Within the narrower discipline of computational stylometry and AI text forensics, the shortfall breaks down into four concrete gaps:

- Gap 1 : Grok is absent from every public attribution benchmark, including RAID and M4, despite occupying a usage share comparable to the other three commercial models by late 2025.
- Gap 2 : every rewriting robustness study published so far measures binary detection degradation. None measures attribution degradation: whether a 4 class classifier can still name which model wrote a text once that text has been rewritten.
- Gap 3: the rewriting tools used in the existing literature, principally DIPPER and similar T5 based paraphrasers, operate through word-level synonym substitution. They do not reflect how a student actually rewrites AI output in 2025, which typically means prompting a second instruction-following LLM to rewrite the text “in my own voice.”
- Gap 4: no published study has examined which individual stylometric or semantic features survive rewriting and which collapse first, information that would be directly useful for designing the next generation of attribution tools.

Taken together, these four gaps mark out a research space the existing literature has not entered: inter-model attribution, across four commercial models including Grok, measured before and after realistic LLM based rewriting, with a feature level account of what survives and what does not. Closing that combined gap, rather than any one piece of it in isolation, is what this study sets out to do.

1.4 Research Question

This study is organised around a single overarching research question, expressed in Wh form, from which four supporting sub questions are derived.

Main Research Question: *How do LLM stylometric fingerprints survive rewriting pressure, and which attribution features resist erasure across ChatGPT, Claude, Gemini, and Grok under LLM based rewriting at light and heavy intensity levels?*

Four sub-questions operationalise this main question into testable parts:

RQ1: What is the baseline inter model attribution accuracy across ChatGPT, Claude, Gemini, and Grok, using a combined stylometric and SentenceTransformer semantic feature set reduced via PCA to approximately 83 dimensions, evaluated through 5-fold stratified cross-validation?

RQ2: How does LLM-based rewriting affect attribution accuracy at light (~40% word-level edit) and heavy (~65% word-level edit) intensity levels, and at what point does attribution accuracy fall below the 25% chance baseline for a 4 class problem?

RQ3: Which of the eight stylometric features retain the most discriminative power under rewriting pressure, as measured through SHAP TreeExplainer survival analysis, and does aggregate semantic signal survive rewriting at a rate comparable to stylometric signal?

RQ4: Does a combined stylometric-semantic feature set outperform either feature type used alone, holding XGBoost constant as the classifier so that the feature type effect is isolated from any classifier architecture effect?

1.5 Research Motivation

The starting point for this project was not one dramatic discovery but an accumulation of small irritations while working through the AI detection literature during the early proposal stage. Krishna et al.'s (2023) DIPPER paper was the first to genuinely unsettle the researcher's assumptions: a single paraphrasing pass collapsing DetectGPT from 70.3% accuracy to 4.6% is not a minor robustness gap it reads more like evidence that the entire detection paradigm rests on much shakier ground than most

institutions appear to assume. Reading the RAID paper (Dugan et al., 2024) shortly afterward reinforced the same impression at a larger scale, across eleven detectors and eleven attack types.

What kept nagging at the researcher, though, was a far simpler observation: Grok was nowhere in either paper, nor in M4 (Wang et al., 2024), nor in any other benchmark turned up during the literature search. By the second half of 2025, Grok had become a default option inside X for hundreds of millions of users, yet not one academic attribution study had included it. That absence felt less like an oversight and more like a structural blind spot that any forensic tool built today would simply inherit.

A second, more technical observation followed naturally from the first. Every rewriting study the researcher could locate measured whether a paraphrased text could still be flagged as AI generated at all. None asked whether the more specific, more forensically useful signal which model wrote it survives the same pressure. Once that distinction became clear, in discussion with the project supervisor, Prof. Chaminda, the direction of the study more or less set itself. An estimated total API cost of roughly \$0.40 made the project realistic to execute single handedly within the available timeline, which made committing to the direction considerably easier.

1.6 Research Aim

The aim of this study is to empirically measure the extent to which inter model attribution accuracy survives LLM based rewriting pressure across four major commercial language models ChatGPT, Claude, Gemini, and Grok and to identify which specific stylometric and semantic features remain discriminative once that rewriting has taken place.

Meeting this aim means producing three concrete deliverables by the end of the project: a measured attribution accuracy curve across the original, lightly rewritten, and heavily rewritten conditions; a ranked account of which of the eight stylometric features, and which semantic signal block, survive that curve best; and a working attribution pipeline that later researchers can extend to additional models, rewriting tools, or languages without redesigning the underlying methodology.

1.7 Research Objectives

Four research specific objectives operationalise the aim above, following the identify analyse design evaluate structure expected at undergraduate level.

1.7.1 To Identify

To identify the stylometric and semantic features that differentiate AI generated text across ChatGPT, Claude, Gemini, and Grok under three edit conditions original, lightly rewritten, and heavily rewritten.

1.7.2 To Analyze

To analyze how LLM based rewriting at light (~40%) and heavy (~65%) intensity levels degrades inter model attribution accuracy across the four model classes, relative to the 25% chance baseline for a 4 class problem.

1.7.3 To Design and Develop

To design and develop a PCA corrected soft-voting ensemble attribution classifier combining eight stylometric features with all MiniLM-L6-v2 SentenceTransformer embeddings, trained exclusively on original samples and evaluated out of distribution on rewritten text without retraining.

1.7.4 To Evaluate

To evaluate which specific stylometric features, and whether aggregate semantic signal, resist LLM based rewriting pressure, using SHAP TreeExplainer survival analysis at both the light and heavy edit intensity levels.

1.8 Rich Picture of the Proposed Solution

Figure 1.1 sets out the complete workflow this study implements, from raw data collection through to the feature-survival findings that answer RQ1 through RQ4. The picture is organised around one structural decision that defines the whole project: the dataset forks into two parallel paths immediately after collection one that stays untouched and one that passes through LLM based rewriting and the gap between what each path produces at the evaluation stage is the actual measurement this thesis reports.

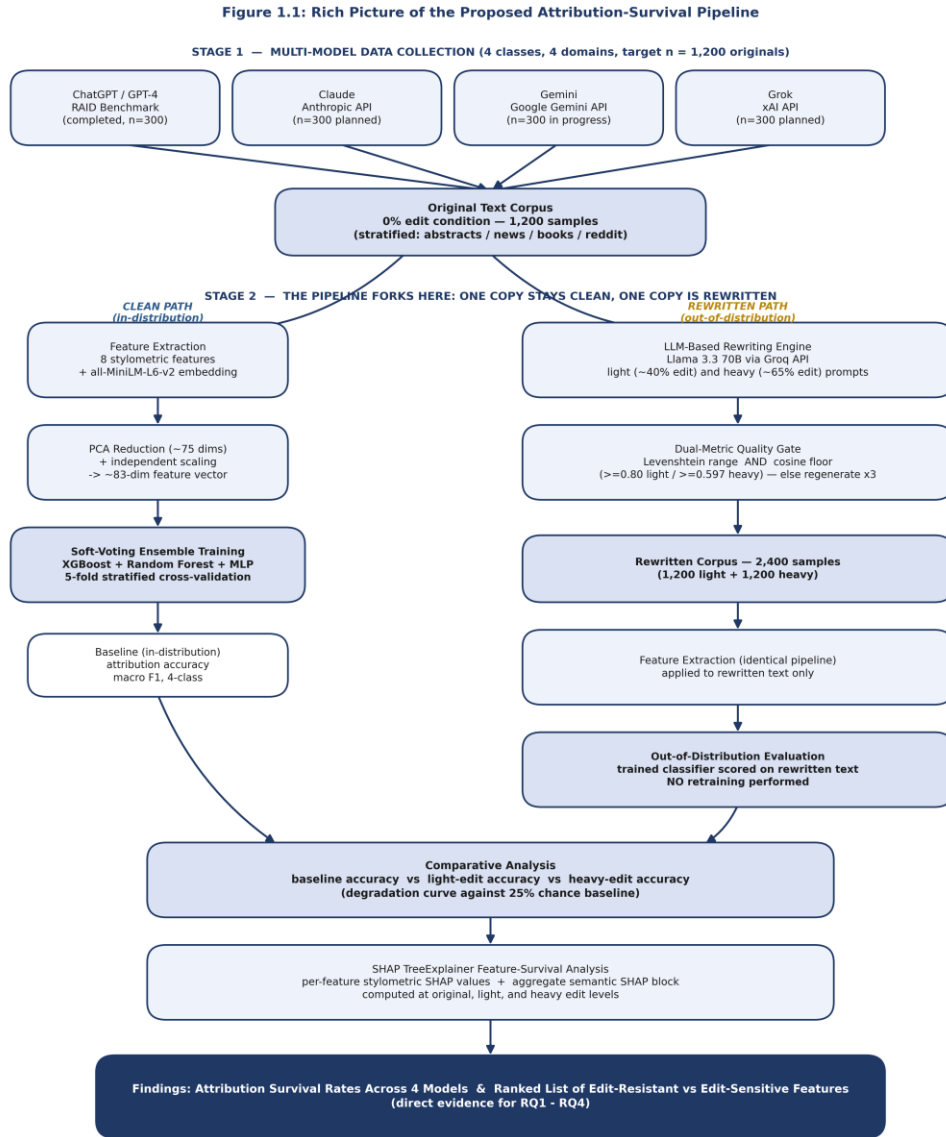


Figure 1.1: Rich picture of the proposed attribution survival pipeline

The pipeline begins with four parallel data-collection streams (Stage 1 in Figure 1.1). ChatGPT and GPT 4 samples come from the RAID benchmark and have already been extracted 300 samples each, filtered to attack='none' and decoding='greedy' conditions. Claude, Gemini, and Grok samples are collected directly from their respective provider APIs, since none of the three appears in any existing public benchmark in the format this study requires. All four streams merge into a single, domain stratified corpus of 1,200 original samples 300 per model, split evenly across academic abstracts, news reporting, creative book style prose, and informal Reddit style text.

From that point, the pipeline forks. On the clean path, the original samples pass straight through feature extraction eight hand engineered stylometric features plus a 384 dimensional all MiniLM-L6-v2 sentence embedding followed by PCA reduction to roughly 75 components and independent feature scaling, producing an approximately 83 dimensional vector per sample. That vector trains the primary classifier: a soft voting ensemble of XGBoost, Random Forest, and a multi layer perceptron, evaluated through 5 fold stratified cross validation. This produces the baseline, in distribution attribution accuracy that anchors every later comparison.

On the rewritten path, the same 1,200 original samples are passed through Llama 3.3 70B via the Groq API under two separate instruction prompts a light rewrite targeting roughly 40% word level change, and a heavy rewrite targeting roughly 65% change. Every rewritten sample must clear a dual quality gate before being accepted: a Levenshtein distance band confirming that enough surface level editing has actually occurred, and a cosine similarity floor (≥ 0.80 light, ≥ 0.597 heavy) confirming that the rewrite has not drifted away from the original meaning. Samples that fail are regenerated up to three times before being replaced from a 50 sample reserve pool. The result is a 2,400 sample rewritten corpus, which passes through the identical feature-extraction pipeline used on the clean path and is then scored by the already trained classifier with no retraining the standard out of distribution evaluation protocol.

The two paths converge at the comparison stage. Baseline accuracy, light edit accuracy, and heavy edit accuracy are placed side by side against the 25% chance baseline, producing the central degradation curve this study reports. SHAP TreeExplainer is then applied to the XGBoost component of the ensemble at all three edit levels, generating individual SHAP values for each of the eight stylometric features and a single aggregate SHAP value for the semantic block as a whole. That feature survival ranking, together with the degradation curve, forms the direct evidence used to answer RQ1 through RQ4.

1.9 Resource Requirements

The technical design of this study was shaped from the outset by the resources realistically available to a self funded undergraduate researcher. Every tool listed below is either already owned, freely available, or accessible at negligible per sample cost, which kept the entire data collection budget under one US dollar.

1.9.1 Hardware

Resource	Specification	Role in the Study
GPU	4 GB VRAM	Runs all MiniLM-L6-v2 for semantic embedding extraction only; the model itself is roughly 80 MB, so 4 GB is more than sufficient.
RAM	24 GB	Holds the full feature matrices, the 3,600 sample dataset, and the soft voting ensemble during training and cross-validation without swapping.
CPU	6 cores / 12 threads	Parallelises stylometric feature extraction, the 5-fold cross validation loop, and SHAP value computation across folds.
Storage	Standard SSD, < 200 MB required	The entire dataset of 3,600 samples, together with extracted embeddings and intermediate feature files, occupies well under 200 MB.

1.9.2 Software

Category	Tool / Library	Purpose
Data collection	HuggingFace datasets (RAID)	Extraction of ChatGPT and GPT 4 samples using the liamdugan/raid identifier.
Data collection	Anthropic API, Google Gemini API, xAI API	Direct generation of Claude, Gemini, and Grok samples, since none appear in RAID.
Rewriting	Groq API (Llama 3.3 70B)	LLM based rewriting of all 2,400 light and heavy edit samples.

Category	Tool / Library	Purpose
Feature extraction	spaCy (en_core_web_sm), NLTK, textstat	Sentence parsing, tokenisation, hedging word counts, TTR, and Flesch Kincaid readability.
Feature extraction	SentenceTransformers (all MiniLM-L6-v2)	384-dimensional semantic embeddings, reused from the quality gate pipeline.
Machine learning	scikit-learn, XGBoost, SHAP	Random Forest, MLP, PCA, StandardScaler, the soft-voting ensemble, and TreeExplainer-based feature survival analysis.
Analysis & reporting	pandas, NumPy, Matplotlib, Seaborn	Data wrangling, descriptive statistics, and all result visualisations.

Total direct project cost is estimated at approximately \$0.40 around \$0.20 for Claude sample generation via the Anthropic API and \$0.20 for Grok sample generation via the xAI API. RAID extraction, the Google Gemini free tier, the Groq API, and every Python library listed above carry no cost to the researcher.

1.10 Project Scope

Table 1.1 sets out the explicit boundaries of this study. Several of these boundaries were tightened deliberately, not because the excluded work is unimportant, but because each one represents a distinct experimental design problem that a single undergraduate project, working to a September 2026 submission deadline, cannot also absorb.

In Scope	Out of Scope
Four commercial LLMs as generation sources: ChatGPT, Claude, Gemini, and Grok.	Any LLM beyond these four (e.g., LLaMA, Mistral, DeepSeek) used as a generation source for attribution.

In Scope	Out of Scope
LLM-based rewriting via Llama 3.3 70B at light (~40%) and heavy (~65%) word-level edit intensity.	Commercial AI humanisation tools such as StealthWriter or Undetectable.ai, and statistical evasion techniques such as perplexity injection.
English language text across 4 domains: academic abstracts, news, creative/book prose, and informal Reddit style writing.	Cross lingual or multilingual attribution.
A combined feature representation: 8 stylometric features plus PCA reduced SentenceTransformer embeddings.	End to end deep learning attribution classifiers (e.g., a fine tuned transformer) as an alternative to the ensemble design.
A soft voting ensemble (XGBoost, Random Forest, MLP) trained on original samples and evaluated out-of-distribution on rewritten samples.	Retraining or fine tuning the classifier directly on rewritten samples.
SHAP TreeExplainer feature-survival analysis on the XGBoost component, at the individual stylometric level and the aggregate semantic level.	Real-time deployment, productisation, or integration into an institutional plagiarism detection system.
RAID benchmark extraction for ChatGPT/GPT 4; direct API collection for Claude, Gemini, and Grok.	Stylometric behaviour of GPT 4o or later model revisions, which are not represented in the RAID benchmark snapshot used here.

Each excluded item is excluded for a specific, statable reason rather than by default. Commercial humanisation tools were left out because they apply additional statistical manipulation perplexity injection and burstiness modification that would require a different quality-gate design entirely; testing their effect on attribution is a natural next study, not a gap in this one. Cross lingual attribution was left out because the RAID benchmark, the primary ChatGPT data source, is English only, and introducing a

second language would multiply the data collection burden well beyond what the timeline allows. The GPT4o boundary is acknowledged directly rather than glossed over: RAID's ChatGPT samples reflect GPT 3.5 and GPT 4 generation configurations from 2023 to 2024, and stylometric fingerprints may have shifted in later model revisions. That is treated here as a documented dataset boundary, not a methodological flaw to be hidden.

1.11 Chapter Summary

This chapter opened with what is actually happening on university campuses in 2025 and 2026: four major commercial LLMs in everyday use, detection tools built around a binary question that no longer matches student behaviour, and a rewriting step that may be quietly erasing whatever forensic signal those tools still capture. From that background, the chapter built a problem statement grounded in measured figures rather than assumption, named four specific research gaps, and set a single Wh form research question with four supporting sub questions. It explained the motivation behind choosing this particular gap, stated the research aim, and broke that aim into four research specific objectives. The rich picture in Section 1.8 then walked through exactly how data moves from four collection sources, through a fork into clean and rewritten paths, into a shared classifier, and out the other side as a degradation curve and a feature survival ranking. The chapter closed with the concrete hardware and software resources committed to the project and an explicit statement of what this study does, and does not, attempt to cover. Chapter 2 now turns to the existing literature that this problem statement was built against.

CHAPTER 2

LITERATURE REVIEW

2.1 Chapter Overview

This chapter builds the theoretical and empirical foundation for the research design. Five interconnected bodies of literature are examined: the current state of AI generated text detection, existing systems and frameworks for attribution and detection, the algorithms that underpin them, the design and workflow choices researchers have made, and the technical tools this study draws on. Each section moves progressively from broad context toward the specific technical gap this study fills. Section 2.6 closes the chapter by linking the findings from each literature area to the four gaps identified in Chapter 1 and offering a critical personal reflection on why existing work, despite its volume, has not answered the research question this study poses.

2.2 Conceptual Map of the Literature

Figure 2.1 below presents the conceptual map of the five literature streams reviewed in this chapter and the logical relationships between them. The domain overview (Section 2.3) provides the broad context within which detection and attribution sit. Existing systems (Section 2.4) are assessed comparatively to identify what has and has not been attempted before this study. The technological analysis (Section 2.5) drills into the algorithmic, design, and workflow choices that distinguish approaches from one another. The reflection (Section 2.6) integrates findings across all five streams into a consolidated gap statement.

Figure 2.1: Conceptual Map of Literature Streams

DOMAIN OVERVIEW (§2.3) AI text detection landscape Binary detection → Attribution Commercial LLM proliferation		EXISTING SYSTEMS (§2.4) RAID Benchmark M4 Benchmark DetectGPT OR GPTZero Styleo metric frameworks
	↓ feeds into ↓ TECHNOLOGICAL ANALYSIS (§2.5) Algorithmic · Design · Workflow ↓ converges in ↓	

TECHNICAL TOOLS SentenceTransformers XGBoost / RF / MLP SHAP / PCA		RESEARCH GAP (§2.6 Reflection) Grok absent No attribution degradation data T5 vs LLM rewriting No SHAP survival analysis
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2.3 Domain Overview

The emergence of large language models as public-facing tools has quietly restructured what academic integrity enforcement means in practice. Until 2022, the question institutions needed to answer was relatively contained: did a student copy from another human source? Today that question has a second dimension did a student use an AI tool and a third that barely anyone has addressed yet which AI tool did they use, and does the evidence of that usage survive if they tried to disguise it?

The field of AI generated text detection developed in direct response to GPT 2's public release in 2019 (Solaiman et al., 2019). What began as a narrow binary classification problem has since expanded into a technically rich research area spanning statistical detection, watermarking, neural classification, and most recently, multi class attribution. By early 2025, four dominant commercial LLMs ChatGPT, Claude, Gemini, and Grok are used by millions of students globally. Padhi et al. (2025) documented distinct adoption patterns across these four models, noting that Grok users skew toward shorter, opinion-driven prompts while ChatGPT users generate longer, more academically structured outputs. These behavioural differences matter for attribution: they imply that stylometric fingerprints vary not just because of how models are built, but because of how users interact with them.

A crucial distinction that the detection literature has historically blurred is the difference between detecting that text is AI generated and attributing it to a specific source model. Binary detection asks a yes or no question. Attribution asks a four-class question. These are not equivalent in difficulty, in the features required, or in the institutional value of the answer. Knowing that a submission is AI generated triggers a process. Knowing which model generated it and knowing that the student's 2nd year submission used ChatGPT while the 3rd year submission used Grok builds a chain of evidence that binary detection cannot provide. That distinction is the domain level motivation for this entire study(4 level).

2.4 Existing Systems, Frameworks, and Designs

Five existing systems and frameworks are most directly relevant to this study. Each is assessed below both for what it contributes and for where it falls short relative to the specific research problem at hand.

2.4.1 RAID Benchmark

The RAID benchmark, introduced by Dugan et al. (2024) at ACL, is the most comprehensive evaluation framework for AI text detection tools to date. It covers eleven language models across eight text domains and eleven adversarial attack types including synonym substitution, homoglyph insertion, and paragraph level paraphrasing via the DIPPER model. Every tested detector showed vulnerability under adversarial conditions, with homoglyph attacks causing average accuracy losses of 40.6% across five detectors, though GPTZero proved considerably more resilient at only 0.3% loss under the same attack. Paraphrasing attacks degraded all detectors consistently.

RAID is genuinely valuable, and this study could not exist without it the ChatGPT and GPT 4 samples extracted from RAID form the foundation of the dataset constructed here. That said, RAID has two limitations that matter directly for this work. First, it frames detection as a binary human VS AI task and does not address attribution. Second, it does not include Grok in its model set. On the question of suitability: RAID's infrastructure is not suitable for the present study as an evaluation benchmark, but it is highly suitable as a data source, which is exactly how it is used.

2.4.2 M4 Benchmark

The M4 benchmark (Wang et al., 2024) extended AI text detection to a multi generator, multi-domain, and multi lingual setting. Covering ChatGPT, Claude, Bloomz, LLaMA, and Dolly across several languages and domains, M4 represented a meaningful expansion of the evaluation landscape. Its framing, however, remained binary, and Grok was absent from its model set. M4 was assessed as a candidate data source for this study and ruled out, it lacks Claude and Gemini coverage in the format required for stylometric attribution at the token level, and its annotation metadata does not include the domain stratification structure needed for the balanced experimental design. M4 is suitable for binary multi lingual detection research. It is not suitable as a data source or evaluation benchmark for this study.

2.4.3 DetectGPT and GPTZero

DetectGPT (Mitchell et al., 2023) is a 0 shot binary detector that exploits the property that LLM generated text tends to occupy local maxima in the log-probability landscape of the generating model small perturbations consistently lower its probability, whereas human writing does not show this property. It is conceptually elegant but has a fundamental deployment constraint: it requires access to the log probability function of the suspected source model, which is not available for any closed commercial API. At scale, this makes DetectGPT unsuitable for institutional deployment.

GPTZero uses perplexity and burstiness as detection signals and does not require source model access. It proved notably robust under homoglyph attacks in the RAID evaluation, losing only 0.3% accuracy. However, GPTZero is a binary classifier that does not distinguish among model sources, and its design is opaque (it is a commercial product with no published architectural details). Neither DetectGPT nor GPTZero is suitable for inter-model attribution, which requires multi class probability outputs and interpretable feature importance scores. They are reviewed here for completeness and to establish the binary detection baseline against which this study's attribution design is distinguished.

2.4.4 Stylometric Attribution Frameworks

Argasinski et al. (2025) demonstrated that stylometric features function word distributions, character n grams, sentence length statistics, and punctuation patterns can distinguish among LLM generated texts even in samples as short as 100 words, using tree based classifiers. This is the closest existing work to the design proposed in this study. The critical gap: Argasinski et al. (2025) tested on unmodified text only. No rewriting condition was included. Whether the stylometric fingerprints they identified survive a single pass of LLM based rewriting at 40% or 65% edit intensity is entirely unknown from their work.

Przystalski et al. (2025) used the StyloMetrix framework for similar purposes, achieving reliable separation of human and LLM generated short texts using lexical, syntactic, and punctuation features. Again, no rewriting condition was tested, and Grok was absent from the model set. Both studies are assessed as directly relevant to the algorithm design choices in Section 3.6 of this study, but neither constitutes a suitable framework for the present research question as stated.

Table 2.1: Comparative Assessment of Existing Systems

System/Study	Grok Included?	Attribution?	Rewriting Tested?	Interpretability?	Suitability for Present Study
RAID Dugan et al. (2024)	No	No	Partial (DIPPER)	No	Data source only, not evaluation benchmark
M4 Wang et al. (2024)	No	No	No	No	Not suitable, missing Claude/Gemini domains
DetectGPT Mitchell et al. (2023)	No	No	No	No	Not suitable, requires source model access
GPTZero	No	No	No	No	Not suitable, binary only, opaque design
Argasinski et al. (2025)	No	Partial	No	Partial	Algorithm design reference , not data source
THIS STUDY	YES	YES	YES LLM	SHAP	Designed to close all gaps simultaneously

The most striking observation from Table 2.1 is not any individual limitation it is the pattern. Every existing system reviewed scores 'No' on at least three of the five criteria. This is not a coincidence or an oversight by individual researchers. It reflects the fact that the four gaps addressed in this study have simply not been prioritised as a combined research question before now. The field has matured along the binary detection axis while the attribution axis, and particularly the attribution under rewriting axis(two way), has been left largely unexamined.

2.5 Technological Analysis

2.5.1 Algorithmic Analysis

Three algorithmic families are relevant to this study: statistical detection algorithms, stylometric attribution algorithms, and ensemble classification methods.

Statistical detection algorithms DetectGPT's log probability curvature test, watermarking (Kirchenbauer et al., 2023), and perplexity thresholding all target the binary human VS AI question.

They are reviewed here because this study's rewriting evaluation is conceptually analogous to the robustness tests applied to these algorithms in prior work: the question of whether a signal survives perturbation is structurally the same whether the signal is a watermark, a perplexity profile, or a stylometric fingerprint. Krishna et al. (2023) established that DIPPER paraphrasing reduces DetectGPT accuracy from 70.3% to 4.6% a finding that directly motivated asking the same robustness question in an attribution context. That translation from binary detection robustness to attribution robustness is the core algorithmic contribution of this study.

Stylometric attribution algorithms operate on extractable, interpretable writing features rather than model internal statistics. The specific features most reliably discriminating between commercial LLMs, according to Argasinski et al. (2025), include function word frequency distributions, sentence length variance, punctuation density, and passive voice rates. Przystalski et al. (2025) found that syntactic complexity measures particularly the ratio of subordinate clauses to total sentences differ significantly across GPT family models and Claude. This study operationalises eight stylometric features drawn from both papers' findings, selecting those with both established discriminative power and interpretable SHAP scores.

Ensemble classification algorithms are used here specifically because the classification target four commercial LLMs whose outputs are stylistically close is a hard multi class problem with a risk of overfitting on the training distribution. Soft voting ensembles combining XGBoost, Random Forest, and MLP offer complementary inductive biases: XGBoost captures non linear feature interactions through boosted trees, Random Forest provides diversity through bootstrap sampling and random feature selection, and MLP captures distributional patterns through non linear transformations of the PCA reduced semantic embedding. Lundberg and Lee (2017) established that SHAP (SHapley Additive exPlanations) provides theoretically grounded feature importance scores for tree-based models; the TreeExplainer variant is used here on the XGBoost component specifically because it is computationally exact for tree structures, unlike the KernelExplainer required for neural models.

The PCA dimensionality reduction applied to the 384 dimensional all MiniLM-L6-v2 embeddings is an algorithmic necessity rather than a choice. Bishop (2006) formalized the curse of dimensionality: when training observations are sparse relative to feature space dimensionality, classifiers model noise rather than signal. With 960 training instances and 392 features without PCA, the ratio of 2.4:1 falls well below the recommended minimum of 5:1 for reliable generalisation. After PCA reduces the semantic block to approximately 75 components, the combined 83 dimensional feature vector achieves a ratio of 11.6:1 within the safe operating range for all three ensemble components.

2.5.2 Design Analysis

The design choices in this study depart from prior attribution work in three specific ways, each motivated by identified limitations in existing designs.

First, the feature design combines stylometric and semantic features rather than using either in isolation. Argasinski et al. (2025) and Przystalski et al. (2025) both found that combining several feature categories lexical, syntactic, and punctuation based produced more reliable separation than relying on any single feature type, a pattern this study extends by combining stylometric features with a semantic embedding block. The rationale is direct: stylometric features capture how sentences are built, while sentence level semantic embeddings capture what kind of content and framing each model tends to produce. These are genuinely complementary information channels. Whether the combination outperforms either type alone in the specific context of this study four commercial LLMs across four text domains is tested empirically in the ablation study rather than assumed from prior findings.

Second, the quality gate design is dual metric rather than threshold based on a single measure. Designs in prior rewriting studies typically use Levenshtein distance alone, which can be satisfied by random word shuffling that destroys meaning, or semantic similarity alone, which can be satisfied by trivial edits that leave the surface form nearly unchanged. The pilot study conducted on 23 February 2026 empirically derived cosine similarity thresholds (0.80 for light, 0.597 for heavy) from human coherence ratings independent of metric values, ensuring the gate reflects actual sample quality rather than an arbitrary numerical cutoff.

Third, the out of distribution evaluation design is used rather than a held out test set from the same distribution. This choice is deliberate. The study's research question is not 'how well does the classifier generalise to unseen original samples' but rather 'how much attribution signal survives rewriting.' Answering the second question requires training on original text and evaluating on rewritten text with no classifier modification. This is the standard OOD robustness evaluation protocol (Hendrycks & Gimpel, 2017; Ovadia et al., 2019) adapted to an attribution context, and it is the only design that produces a directly interpretable measurement of attribution survival.

2.5.3 Workflow Analysis

The workflow designs of existing detection and attribution systems are compared here against the workflow of the present study.

RAID's evaluation workflow is linear: collect samples from multiple generators under multiple adversarial conditions, submit to detectors, measure accuracy. This workflow is well suited to benchmarking existing tools but cannot answer questions about the internal feature level mechanics of why detectors succeed or fail under specific conditions. RAID does not include a feature extraction stage, a classifier training stage, or an interpretability stage. Its workflow is an evaluation pipeline, not a modelling pipeline.

DetectGPT's workflow is entirely inference time: take a candidate text, generate perturbations, query the source model's log probability function for each perturbation, classify based on the curvature of the probability landscape. This workflow requires zero training data and produces a binary output. It has no feature extraction stage and no interpretability mechanism. The workflow is elegant but closed.

The workflow of this study differs from both. It proceeds through six distinct stages: (1) data collection and quality gate filtering, (2) stylometric feature extraction using spaCy, NLTK, and textstat, (3) semantic embedding using all-MiniLM-L6-v2 and PCA reduction, (4) classifier training on original samples via 5-fold stratified cross-validation, (5) OOD robustness evaluation on rewritten samples at two edit levels without retraining, and (6) SHAP feature survival analysis on the XGBoost component. Stages 5 and 6 together constitute the novel contribution of the workflow: no existing attribution pipeline includes both an OOD degradation measurement and a feature-level survival analysis at multiple rewriting intensities.

Table 2.2: Workflow Comparison Existing Systems vs This Study

Workflow Stage	RAID	DetectGPT	Argasinski (2025)	This Study
Data collection with quality control	Partial	No	No	Yes dual metric quality gate
Stylometric feature extraction	No	No	Yes	Yes 8 features (spaCy, NLTK, textstat)
Semantic embedding + PCA reduction	No	No	No	Yes oll MiniLM-L6-v2 → 75 PCA components

Cross-validated classifier training	No	No	Yes	Yes 5 fold stratified, 3 model soft voting ensemble
OOD rewriting robustness evaluation	Partial	No	No	Yes two edit levels, no classifier retraining
Feature survival interpretability (SHAP)	No	No	No	Yes SHAP TreeExplainer on XGBoost component

2.6 Reflection

Having reviewed the domain, existing systems, and the technological choices underpinning them, several observations emerge that go beyond what any single paper states explicitly.

The most striking thing about the attribution literature is not any individual paper's limitation it is the collective direction of attention. Researchers have invested enormous effort in binary detection while treating attribution as an afterthought. From an academic integrity perspective, this ordering is arguably backwards. Knowing that AI was used is the beginning of an investigation, not its conclusion. Knowing which model was used, across multiple submissions, across multiple years, is what builds an evidence based institutional response. The field has been solving the easier problem while the harder and more useful problem sits largely unaddressed.

The rewriting robustness literature compounds this concern. Krishna et al. (2023) showed that DIPPER paraphrasing reduces binary detection accuracy from 70.3% to 4.6% a finding that effectively challenged the value proposition of the entire detection industry at the time. Yet five years later, the natural follow-up question what happens to attribution accuracy under the same pressure has not been answered. Zhang et al. (2025) extended the robustness analysis to adversarial paraphrasing without answering it. Dugan et al. (2024) included paraphrasing as one of eleven attack types in RAID without asking what it does to per model attribution confidence. The gap is not hidden or subtle. It is large, obvious, and practically significant. The fact that it remains open reflects a research culture that has been more comfortable evaluating what already exists than building what does not yet exist.

The Grok gap is a separate issue and, in my assessment, a more embarrassing one. Ye et al. (2025) found that classifiers trained on ChatGPT, Claude, and Gemini achieve zero agreement with Grok-generated outputs. This is not a marginal accuracy drop it is a categorical failure. And yet the benchmark

infrastructure that dominates the field (RAID, M4, DAIGT) was built without Grok and has not been updated to include it. Building an attribution system and benchmarking it without Grok in 2025 is analogous to building an email spam filter and forgetting to include one of the four largest spam networks. The contribution of including Grok in this study is not merely additive it is corrective.

On the technical side, the choice of all MiniLM-L6-v2 over raw BERT CLS representations is worth reflecting on explicitly. Reimers and Gurevych (2019) demonstrated that unfinetuned BERT CLS representations perform worse than averaged word2vec embeddings on sentence similarity tasks a result that was surprising at the time and that has been consistently replicated since. The semantic embedding literature has moved decisively toward contrastively trained SentenceTransformer architectures, yet a number of recent attribution papers continue using raw BERT CLS as their semantic feature. This study uses all MiniLM-L6-v2 precisely because the semantic representation quality difference is not marginal in the context of attributing stylistically similar text. When the classification target is four models that have converged stylistically, every percentage point of embedding quality matters.

The SHAP analysis design also reflects a choice that I consider methodologically important and that is not made by default in the attribution literature. Reporting aggregate feature importance without separating stylometric from semantic contributions obscures the practically useful finding. What institutions and system designers need to know is not just that 'combined features work best' but specifically which stylometric features carry attribution signal through rewriting pressure. That is actionable. 'Combined features outperform individual types' is informative but not actionable. Keeping stylometric features outside the PCA transformation and computing their SHAP values individually rather than folding them into PCA components that carry no direct interpretable meaning is a deliberate design choice in service of that actionable finding.

Taken together, the literature reviewed in this chapter establishes a field that is technically sophisticated, methodologically rigorous within its established question set, and collectively neglectful of the specific question that matters most for institutional AI attribution in 2025. This study is designed to address that neglect directly, systematically, and at a scale appropriate for an undergraduate research project with a clear contribution boundary.

2.7 Chapter Summary

This chapter reviewed five interconnected bodies of literature to build the theoretical and technical foundation for the present study. The domain overview established that AI text detection has matured as a binary classification field while inter-model attribution has been comparatively neglected.

The comparative assessment of existing systems RAID, M4, DetectGPT, GPTZero, and stylometric attribution frameworks found that no existing system includes Grok, none tests attribution accuracy under rewriting pressure, and none combines stylometric and semantic features with a SHAP survival analysis. The technological analysis examined the algorithmic choices (statistical detection, stylometric attribution, ensemble classification, SHAP, PCA), design choices (hybrid features, dual-metric quality gate, OOD evaluation protocol), and workflow structures (data collection, feature extraction, training, OOD evaluation, interpretability) that distinguish this study from prior work. The reflection section offered a critical personal reading of the literature's collective gaps, arguing that the field has prioritised the easier binary detection problem over the harder and more institutionally valuable attribution problem. Four research gaps were confirmed: the Grok gap, the absence of attribution degradation data, the methodological mismatch between T5 based rewriting tools and actual student behaviour, and the lack of SHAP based feature survival analysis. Chapter 3 presents the methodology designed to close all four gaps within a single experimental framework.

CHAPTER 3

METHODOLOGY

3.1 Research Paradigm

Every research project, whether the researcher acknowledges it or not, sits within a philosophical framework that shapes what is treated as real, what counts as knowledge, and how that knowledge is gathered. Making those choices explicit matters particularly in computational research, where the technical machinery of measurement can create an illusion of philosophical neutrality that does not actually exist. This section states the philosophical position of the present study clearly.

The study adopts a realist ontological position. Stylometric fingerprints the statistical regularities in sentence structure, vocabulary distribution, punctuation frequency, and syntactic complexity that distinguish one LLM's output from another's are treated as objective properties of the generated text that exist independently of the researcher's observation. They are measurable, reproducible, and consistent across samples drawn from the same model under the same conditions. This is not a trivial assumption: a constructivist would argue that 'stylometric fingerprint' is a category imposed by the researcher rather than a property of the text itself. The realist position is taken here because the evidence from Argasinski et al. (2025), Przystalski et al. (2025), and the preliminary RAID extraction confirms that these regularities are stable enough to support cross-validated classification at above-chance accuracy.

Epistemologically, the study is empiricist. Knowledge about attribution accuracy and feature survival is derived entirely from systematic observation and measurement training a classifier, applying it to new data, computing accuracy, and computing SHAP importance values. No claims are made about the internal mechanisms of the LLMs being studied, which are closed commercial systems whose architecture details are not publicly available. What can be known is what can be measured from outputs alone.

The broader philosophical stance is post positivist, tilting toward pragmatism where choices had to be made between theoretical purity and practical tractability. A strict positivist would insist that pilot thresholds be derived from a distribution that perfectly matches the target data which would require collecting target data before setting the quality gate, a circular dependency. The pragmatic resolution was to derive thresholds from a closely analogous Llama rewriting Llama scenario, document the limitation

explicitly, and plan a calibration refinement as the first production step. This kind of transparent practical compromise is characteristic of post positivist applied research.

Table 3.1: Research Paradigm Summary

Dimension	Position Adopted	Justification
Ontology	Realist LLM stylometric fingerprints exist as measurable, objective properties of generated text	Attribution signals are real patterns in data, not constructions of the observer
Epistemology	Empiricist knowledge is derived from systematic observation, measurement, and controlled experimentation	Attribution accuracy and feature survival are measured, not inferred from theory alone
Research Philosophy	Post positivist or Pragmatist objective measurement is pursued while acknowledging the limitations of controlled computational experiments	The study measures real phenomena but recognises that pilot derived thresholds and controlled rewriting conditions are approximations of real world behaviour
Axiology	Value laden but transparent the study has an explicit defensive motivation (protecting attribution system integrity)	All design choices, thresholds, and limitations are reported with full transparency to allow independent replication

3.2 Research Approach

The study is primarily deductive in its approach. It begins with established theoretical claims that LLMs produce measurable stylometric fingerprints (Argasinski et al., 2025), that paraphrasing degrades detection accuracy (Krishna et al., 2023), and that attribution classifiers degrade under distribution shift (Hendrycks & Gimpel, 2017) and tests whether these hold in the specific and previously untested context of inter model attribution under LLM based rewriting. The four research questions are stated in advance, the experimental design is fixed before data collection begins, and the evaluation metrics are specified before results are observed. These are the hallmarks of deductive hypothesis testing.

That said, the SHAP survival analysis component introduces a genuinely inductive element. While the study predicts that some features will prove more edit resistant than others based on the intuition that structural features like comma density are harder to alter through rewriting than vocabulary level features like TTR it does not predict which specific features will rank highest or in what order. The SHAP rankings emerge from the data rather than being derived from a prior theoretical ordering. The study is therefore better described as primarily deductive with a structured inductive component at the interpretability stage, which is a common and well established design in applied machine learning research.

A purely inductive approach collecting data first and building theory from whatever patterns emerge would be unsuitable here. This study has a specific and narrow question. Inductive designs are appropriate when the research territory is genuinely unmapped. Here the territory is partially mapped: we know binary detection degrades under rewriting, we know stylometric features can support attribution, and we know Grok is absent from benchmarks. The study is filling a defined gap, not exploring an undefined space. Deductive design with a fixed evaluation protocol is the appropriate methodological choice.

3.3 Research Strategy

The research strategy is controlled computational experimentation. Three edit conditions (0%, ~40%, ~65%) are applied to text from four model classes under systematically controlled conditions same domains, same sample sizes, same prompts across models, same rewriting model, same quality gate. The classifier is trained on one condition and evaluated on the other two without modification. This is a factorial experimental design with one training condition and two test conditions, producing a clean measurement of how much attribution signal survives each level of rewriting pressure.

The strategy differs from a survey strategy (which would gather opinions or existing data without experimental manipulation), a case study strategy (which would examine one model or one rewriting event in depth), and an archival strategy (which would analyse existing datasets without generating new conditions). All three were considered and set aside. Surveys do not produce the kind of quantitative feature level data that the SHAP analysis requires. Case studies sacrifice generalisability across the four model classes. Archival analysis cannot produce the rewriting conditions needed because no existing dataset contains the same texts under both original and LLM rewritten conditions with quality gate filtering.

The computational experimental strategy is appropriate because the research question is causal in structure: does rewriting cause attribution accuracy to drop, and if so, which features drive that drop? Answering a causal question requires manipulating the independent variable (rewriting intensity) while controlling everything else (model classes, domains, sample sizes, prompts). The experimental strategy is the only design that achieves this control.

3.4 Fact Collection Mechanisms

Data is gathered through four distinct mechanisms, each targeting a different component of the total dataset.

The first mechanism is secondary data extraction from the RAID benchmark. ChatGPT and GPT 4 samples 300 each are extracted from the publicly available RAID dataset (liamdugan/raid on HuggingFace) using filtering conditions of attack='none' and decoding='greedy' to ensure clean, unmanipulated text. Domain stratification at 75 samples per domain is applied using a fixed random seed of 42. This extraction has been completed. Files are stored locally as chatgpt_300samples.csv and gpt4_300samples.csv.

The second mechanism is primary data generation via commercial APIs. Claude samples will be collected through the Anthropic API using the Claude Haiku model. Gemini samples are currently being collected via the Google Gemini API (gemini 2.5 flash model), with a billing upgrade planned to resolve the rate limiting issues encountered on the free tier. Grok samples will be collected via the xAI API using whichever Grok model is current at the time of collection, since xAI updates its model line up frequently. For all three API collected models, the same domain-stratified prompts extracted from the RAID samples are submitted directly, ensuring prompt comparability across all four models.

The third mechanism is controlled text transformation via Llama 3.3 70B through the Groq API. This is the rewriting stage: each of the 1,200 original samples is submitted to Llama 3.3 70B under the light and heavy instruction prompts defined in Section 3.5.5, producing 2,400 rewritten samples that then pass through the quality gate before being accepted into the dataset.

The fourth mechanism is automated feature extraction applied uniformly across all 3,600 samples. Stylometric features are extracted using spaCy (v3.x, en_core_web_sm pipeline), NLTK, the textstat library, and standard Python string methods. Semantic embeddings are extracted using the SentenceTransformers library with the all MiniLM-L6-v2 model. These are computational fact collection

mechanisms rather than human data collection mechanisms no human judgement enters the feature extraction process, ensuring full reproducibility.

Table 3.2: Dataset Collection — Sources, Status, and Mechanisms

Model	Source	Status	Notes
ChatGPT	RAID Benchmark (liamdugan/raid)	Done 300 samples	attack=none, decoding=greedy, seed=42. Stored: chatgpt_300samples.csv
GPT-4	RAID Benchmark	Done 300 samples	Same filtering conditions. Stored: gpt4_300samples.csv
Gemini	Google Gemini API (gemini-2.5-flash)	In Progress	Rate limiting on free tier. Billing upgrade planned. Same domain stratified prompts as RAID samples.
Claude	Anthropic API (~\$0.20)	Planned	RAID does not contain Claude samples. Official API required. Claude Haiku model.
Grok	xAI API (~\$0.20)	Planned	Absent from all public benchmarks. Primary novel data contribution. Current Grok model via xAI API (version confirmed at collection time).

3.5 Methodology Execution Workflow

Table 3.3 presents the 6 phase methodology execution workflow. Each phase is described in the subsections below.

Figure 3.1: Methodology Pipeline (End to End Data Flow from presentation)

Methodology Pipeline

Data flows left to right through 8 stages — icons identify each process at a glance

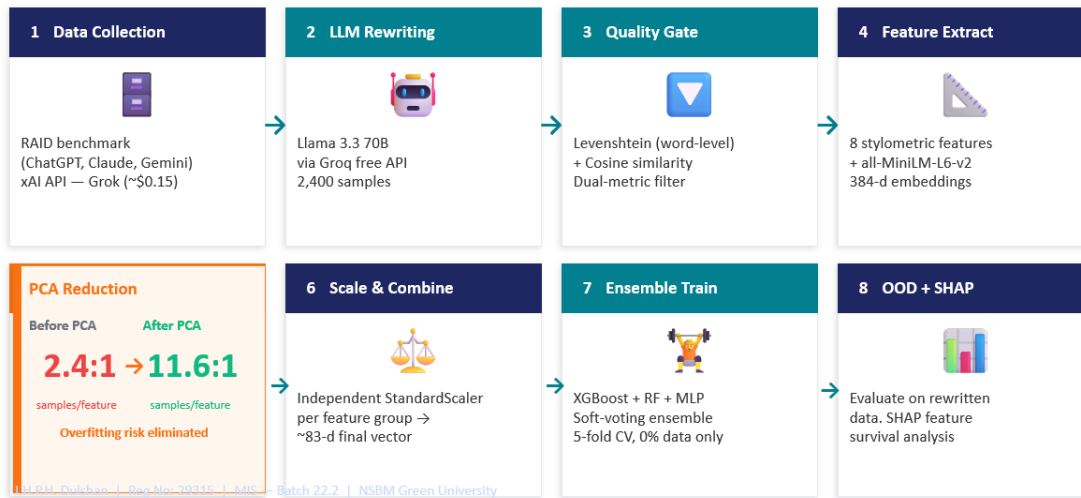


Table 3.3: Methodology Execution Workflow

#	Phase	Activities Undertaken	Output / Deliverable
3.5.1	Problem Identification	Review of RAID, M4, and attribution literature. Identification of Grok absence, attribution degradation gap, and T5 rewriting mismatch with student behaviour in 2025.	Four research gaps. One main research question and four sub-questions.
3.5.2	Relevance Justification	Literature review confirming no prior study measures attribution degradation under LLM based rewriting. Statistical confirmation of Grok's zero agreement problem (Ye et al., 2025).	Structured gap table. Comparative system assessment (Table 2.1). Confirmation of study novelty.
3.5.3	Comparative Analysis and Gap Justification	Comparative assessment of RAID, M4, DetectGPT, GPTZero, and stylometric frameworks. Workflow comparison table produced. Personal	Tables 2.1 and 2.2 in Chapter 2. Confirmed research gap across all four dimensions.

		reflections on field priorities documented.	
3.5.4	Define and Finalise Objectives	Research aim, main RQ, four sub questions, and four objectives finalised. Scope boundaries defined including exclusion of commercial humanisation tools and cross-lingual text.	Research objectives 1.7.1 or 1.7.4. Scope table (Chapter 1). Ethics clearance submitted.
3.5.5	Design, Development, and Data Management	Pilot study conducted 23 Feb 2026. Quality gate thresholds derived. Dataset collection from RAID (completed for ChatGPT/GPT-4) and APIs (Gemini in progress, Claude and Grok planned). Feature extraction pipeline designed. PCA ratio analysis. Classifier and ablation design finalised.	chatgpt_300samples.csv and gpt4_300samples.csv (done). Gemini and Grok collection scripts ready. Feature extraction pipeline. 83-dimensional feature vector specification.
3.5.6	Evaluation and Communication	5-fold stratified CV on original samples. OOD evaluation on both rewriting conditions. Ablation study across 4 configurations. SHAP TreeExplainer survival analysis. Results reported against four RQs.	Macro F1 scores per condition. Confusion matrices. SHAP feature rankings. Thesis chapters 4 and 5.

3.5.1 Problem Identification

The problem was identified through a systematic review of the AI text detection literature conducted between November 2025 and January 2026. RAID (Dugan et al., 2024) and M4 (Wang et al., 2024) were reviewed as the two most comprehensive existing benchmarks. Both frame the detection task as binary and both exclude Grok. The rewriting robustness literature principally Krishna et al. (2023) and Zhang et al. (2025) established that binary detection degrades under paraphrasing without ever asking what happens to attribution accuracy under the same pressure. Ye et al. (2025) confirmed that classifiers trained on three commercial LLMs achieve zero agreement with Grok outputs, quantifying the Grok gap explicitly.

From these observations, four gaps were identified: the absence of Grok from all attribution benchmarks, the absence of attribution degradation measurements under rewriting, the methodological mismatch between T5 based rewriting tools and actual 2025 student behaviour, and the absence of feature level survival analysis. These four gaps together define a research problem that is specific enough to be addressed in an undergraduate project, significant enough to warrant publication, and practically motivated by real institutional concerns about academic integrity infrastructure.

3.5.2 Relevance Justification

The relevance of this study is justified on two levels. At the technical level, the Grok gap is not a minor omission it is a categorical failure of existing systems, as Ye et al. (2025) demonstrated. At the practical level, the shift from binary detection to attribution represents a qualitative improvement in what institutions can actually do with AI text analysis results. A binary detector tells an institution that a submission may be AI generated. An attribution classifier tells it which model was likely used, enabling pattern detection across multiple submissions and more robust evidence chains in misconduct proceedings.

The rewriting dimension adds a further layer of practical relevance. If attribution accuracy drops to chance level after a 40% word level edit, then attribution tools as currently designed have no practical value for any submission that has passed through a second LLM. Quantifying that threshold is arguably more useful for institutional decision making than any accuracy figure measured on unmodified text.

3.5.3 Comparative Analysis and Gap Justification

The comparative analysis was conducted in Chapter 2 and produced Tables 2.1 and 2.2. Five existing systems were assessed across five criteria: Grok inclusion, multi class attribution, rewriting robustness testing, interpretability, and workflow completeness. No system scored positively on all five. The pattern was consistent: every system reviewed was designed to answer the binary detection question and was evaluated on clean, unmodified text. The gap is not localised to any single paper or research group it is structural, reflecting the field's collective prioritisation of binary detection over attribution.

3.5.4 Define and Finalise Objectives

The four research objectives were finalised following the comparative analysis, ensuring that each objective directly addresses one or more of the identified gaps. Objective 1.7.1 (To Identify) targets the feature level question which stylometric and semantic features discriminate among the four models. Objective 1.7.2 (To Analyze) targets the degradation question how attribution accuracy changes under

rewriting pressure. Objective 1.7.3 (To Design and Develop) targets the pipeline question building the complete attribution system from data collection through classification. Objective 1.7.4 (To Evaluate) targets the feature survival question SHAP based analysis of which features resist editing.

3.5.5 Design, Development, and Data Management

The complete pipeline design is documented across the following components.

Rewriting procedure: All 2,400 rewritten samples are generated using Llama 3.3 70B via the Groq API. The light rewriting instruction prompts the model to 'Rewrite the following text in a natural academic tone. Change sentence structures and word choices meaningfully, but preserve all factual content and meaning exactly. Do not add new information.' The heavy rewriting instruction prompts it to 'Completely rephrase the following text. Substantially alter sentence structures, vocabulary, and phrasing throughout the entire passage while preserving the core meaning and maintaining academic coherence. The result should read as if written by a different author.' Both prompts are fixed across all four model classes and both edit conditions.

Table 3.4: Dual-Metric Quality Gate Specification

Metric	Light (~40%)	Heavy (~65%)	Computation Method
Normalised Levenshtein distance	0.30 to 0.50	0.55 to 0.70	<code>edit_distance(tokens_A, tokens_B) / max(len(A), len(B))</code> on whitespace tokenised text via python-Levenshtein
Cosine similarity (pilot-derived threshold)	>= 0.80	>= 0.597	All MiniLM-L6-v2 SentenceTransformer embeddings, mean pooling over full sample text

Each rewritten sample must satisfy both conditions in Table 3.4 simultaneously. Samples failing either condition are regenerated up to three times. After three failures, the sample is replaced from a 50 sample reserve pool constructed using the same collection procedures as the main dataset. All replacements are logged.

Table 3.5: Stylometric Feature Set

Feature	Library	Attribution Rationale
Mean sentence length (tokens)	spaCy	LLMs differ in default sentence complexity; GPT 4 tends longer than Claude (Argasinski et al., 2025)
Sentence length variance	spaCy	Variance captures rhythmic variation a fingerprint stable across topics but distinct across models
Type-Token Ratio (TTR)	NLTK	Lexical diversity proxy varies with model training corpus and vocabulary sampling strategy
Hedging word frequency	NLTK word list	Claude demonstrates higher epistemic hedging than GPT family models (Argasinski et al., 2025)
Passive voice rate	spaCy dependency parse	Academic register preference varies by model RLHF fine tuning and instruction alignment
Comma density (per 100 words)	Python str.count()	Punctuation patterns are stable model fingerprints less susceptible to semantic rewriting pressure
Mean word length (characters)	Python	Vocabulary sophistication level differs between models trained on different corpus distributions
Flesch-Kincaid readability score	textstat	Composite score sensitive to both sentence and word length captures model level register preferences

Feature extraction is performed on the text as it would be presented to the classifier in deployment on the rewritten version for rewritten samples, not on the original. The 384 dimensional all MiniLM-L6-v2 embedding is extracted using mean token pooling. PCA is applied within each cross validation fold: fitted on training data only, applied to test data. The number of retained components is determined by a 95% cumulative variance threshold, yielding approximately 75 components in preliminary analysis. Two independent StandardScalers are applied one to the 8 stylometric features, one to the PCA reduced semantic block both fitted on training data only. The final feature vector of approximately 83 dimensions is assembled by concatenating the two scaled blocks.

3.5.6 Evaluation and Communication

The evaluation design uses 5 fold stratified cross validation on the 1,200 original samples for baseline accuracy, followed by out of distribution evaluation on both rewriting conditions using the trained model without modification. The primary metric is macro F1 score, which weights all four model classes equally. Secondary metrics include per-class recall and confusion matrices. The 25% chance baseline for a 4 class problem is the reference point against which degradation severity is judged.

Four ablation configurations are evaluated as described in Table 3.6 below. XGBoost is held constant as the classifier across configurations 1 through 3 to ensure that any accuracy difference reflects a feature type effect. Configuration 4 introduces the full soft voting ensemble to measure the additional benefit of multi classifier diversity.

Table 3.6: Ablation Study Configurations

No.	Classifier	Feature Set	What It Isolates	RQ Addressed
1	XGBoost (constant)	8 stylometric only	Stylometric signal strength alone	RQ4 baseline
2	XGBoost (constant)	~75 PCA semantic only	Semantic signal strength alone	RQ4 baseline
3	XGBoost (constant)	~83 combined (scaled independently)	Feature type complementarity, isolating classifier effect	RQ4 answer
4 PRIMARY	XGBoost + RF + MLP soft vote	~83 combined (scaled independently)	Ensemble diversity benefit over single classifier	RQ1 and RQ2

SHAP TreeExplainer is applied to the XGBoost component of the primary ensemble. Feature importance is computed on the 0% original test fold samples as baseline, then on both rewriting conditions. Stylometric features are reported individually. Semantic PCA components are aggregated into a single importance sum, not interpreted individually, because PCA components carry no direct linguistic

meaning. The top three edit resistant and edit sensitive stylometric features are identified and interpreted in Chapter 5.

Findings will be communicated through Chapters 4 and 5 of the thesis, submitted as Interim Submission 02 in late July 2026, the complete draft on 18 August 2026, and the final thesis on 01 September 2026. All code, feature extraction scripts, and quality gate implementations will be included as supplementary material.

3.6 Project Management Methodology

SCRUM was selected as the project management methodology for this research. Three alternatives were evaluated before this choice was made.

KANBAN was considered first. KANBAN suits continuous-flow work where tasks arrive unpredictably and priorities shift constantly. Research does involve some unpredictability API rate limiting, unexpected data quality issues, supervisor feedback cycles but the overall work structure is not a continuous flow. It has a defined beginning, a sequence of phases that must be completed in order (data collection before feature extraction, feature extraction before training, training before evaluation), and a fixed end date. KANBAN's lack of defined sprints and deliverable boundaries makes it poorly suited to a project with four hard submission deadlines.

PRINCE2 was considered second. PRINCE2 is a heavyweight project management framework designed for large multi team projects with formal governance structures. It requires stage gates, business cases, and structured escalation paths that are disproportionate to a single researcher undergraduate project. The overhead of PRINCE2 documentation would consume time better spent on the research itself.

SCRUM is appropriate for three reasons specific to this project. First, the work naturally divides into time-boxed sprints with defined deliverables pilot study sprint, data collection sprint, rewriting and extraction sprint, training and evaluation sprint, write-up sprint. Second, SCRUM accommodates supervisor review cycles as sprint retrospectives, integrating feedback into the next sprint's planning rather than treating it as an interruption. Third, SCRUM's emphasis on working deliverables at the end of each sprint aligns well with the interim submission structure Sprint 1 ends with the Chapter 1 to 3 draft, Sprint 4 ends with the results, Sprint 6 ends with the final thesis.

3.6.1 Project Timeline

Table 3.7: SCRUM-Based Project Timeline (Seven Sprints)

Sprint	Phase	Key Activities	Duration	Deliverable
1	Literature Review and Research Design	Literature search, gap identification, proposal writing, ethics clearance	Nov 2025 – Jan 2026	Research proposal, ethical clearance form, Chapter 2 draft
2	Pilot Study and Dataset Planning	Pilot rewriting (23 Feb 2026), threshold derivation, RAID extraction, API setup, chatgpt and gpt4 samples collected	Feb 2026 – Mar 2026	Pilot results, quality gate thresholds, chatgpt_300samples.csv, gpt4_300samples.csv
3	Full Dataset Collection	Gemini API collection (billing upgrade), Claude API collection, Grok API collection, threshold calibration on target model outputs	Apr 2026 – May 2026	All four model CSV files. Calibrated quality gate thresholds.
4	Rewriting and Feature Extraction	LLM-based rewriting via Llama 3.3 70B (Groq API), quality gate filtering, stylometric extraction, SentenceTransformer embeddings, PCA fitting	May 2026 – Jun 2026	2,400 rewritten samples passing quality gate. 83 dimensional feature matrix for all 3,600 samples.
5	Model Training and OOD Evaluation	5-fold CV training on 0% condition, ablation configurations 1 to 3, primary ensemble OOD evaluation on 40% and 65% conditions	Jun 2026 – Jul 2026	Baseline F1 scores, OOD F1 scores, ablation results, confusion matrices

6	SHAP Analysis and Write-Up	SHAP TreeExplainer on XGBoost component, feature survival rankings, thesis Chapters 4 and 5 writing, supervisor review cycles	Jul 2026 – Aug 2026	SHAP survival rankings. Complete thesis draft. Interim Submission 02.
7	Final Revision and Submission	Supervisor feedback integration, proofreading, final formatting, viva preparation	Aug – Sep 2026	Final thesis submission by 01 Sep 2026. Viva ready presentation.

Each sprint is structured with a planning session at the start (reviewing the previous sprint's output and defining the current sprint's tasks), daily self check ins to track progress against the sprint goal, and a review at the end to assess what was completed against what was planned. Supervisor meetings serve as sprint reviews, and the action items from each meeting become the backlog items for the following sprint. The timeline builds in one week of buffer per sprint to absorb unexpected delays API outages, rate limiting, supervisor availability gaps without cascading into subsequent phases.

3.6.2 Ethical Considerations

All data used in this study is either publicly available under an open licence or generated by the researcher using legitimate API access under each provider's terms of service. The RAID benchmark (liamdugan/raid on HuggingFace) is released for academic research use. Claude, Gemini, and Grok samples are collected through official APIs under terms of service that permit academic research use. No human participant data is collected or processed at any stage. The NSBM Faculty of Computing Research Ethics Committee application has been submitted and documents this position formally.

The rewriting procedure uses Llama 3.3 70B to simulate behaviour that students already perform using commercially available tools. This study does not introduce a new adversarial capability it measures and documents an existing one. The stated purpose is explicitly defensive: identifying which attribution features are most vulnerable to rewriting pressure is intended to guide the design of more robust future attribution systems, not to provide a recipe for evading them. The features most at risk vocabulary level metrics like TTR are already documented in the open forensic stylometry literature and are not novel attack vectors.

Dual use risk is acknowledged and addressed through publication framing. The findings will be reported in terms of what institutions and system designers should prioritise in next generation attribution tools, not in terms of what editing strategies most effectively evade detection. This framing is not cosmetic it reflects the genuine research motivation, which emerged from observing that no existing study had told institutions whether their attribution tools remain meaningful after a single rewriting pass.

All API keys and credentials are stored as local environment variables on the researcher's device and are excluded from all code files, supplementary materials, and version control repositories. Raw sample text is stored locally and will not be uploaded to any public or cloud platform. De identified sample datasets without API metadata that could be used to reconstruct API access credentials will be made available upon project completion in support of open science principles.

3.7 Chapter Summary

This chapter documented the complete research methodology across six dimensions. The research paradigm is post-positivist with a realist ontology and empiricist epistemology, adopting a pragmatic stance where theoretical purity conflicts with practical tractability. The research approach is primarily deductive testing specific theoretical claims about attribution degradation under rewriting with a structured inductive component at the SHAP feature survival analysis stage. The research strategy is controlled computational experimentation, manipulating edit intensity while controlling model classes, domains, sample sizes, prompts, and evaluation protocol. Fact collection mechanisms span secondary data extraction from RAID, primary data generation via three commercial APIs, controlled text transformation via Llama 3.3 70B, and automated feature extraction via spaCy, NLTK, textstat, and SentenceTransformers.

The methodology execution workflow was presented across six phases: problem identification, relevance justification, comparative analysis and gap justification, objective finalisation, design and development, and evaluation and communication. The quality gate specification, stylometric feature set, ablation study design, and SHAP analysis structure were each documented with full technical detail. SCRUM was selected as the project management methodology and justified against KANBAN and PRINCE2 alternatives, with a seven sprint timeline mapped to the four submission deadlines. Ethical considerations were addressed comprehensively, covering data sources, dual use risk, publication framing, credential management, and open science commitments. Chapter 4 will present the results of the experiments described here.

